

(a) Three-fold motor-level holdout

Sources: 1.5 + 3 kW → Target: 1 kW

Sources: 1 + 3 kW → Target: 1.5 kW

Sources: 1 + 1.5 kW → Target: 3 kW

No target-fault label enters fitting, selection, adaptation, or calibration

(b) Ordered target-health deployment record



(c) Healthy-only motor-balanced scoring and alarm calibration

Scale-free
current features

Per-motor robust
median / MAD

One healthy
covariance / motor

Equal-motor
Log-Euclidean mean

Target-centered
Mahalanobis score

3-s block max +
healthy p-value

20 target healthy calibration blocks → alarm when upper-tail $p \leq 0.05$